AI-Powered User Access Risk Scoring System

Mehmet Birol Ozder

Department of Computer Engineering, Kastamonu University, Kastamonu, Turkey

Abstract

This paper presents a cloud-based risk scoring automation capable of detecting suspicious login attempts against user accounts in real time. Rather than relying on the static rule sets characteristic of traditional systems, the proposed approach employs a dynamic architecture orchestrated through the n8n workflow automation platform. At the core of the system, the OpenAI GPT-4o-mini large language model analyzes each user's historical location, IP address, and device data to produce a risk score on a scale of 0 to 100. In a controlled test scenario, an attempt by an unknown Android device originating from a Chinese IP address to access an account ordinarily associated with Kastamonu, Turkey was simulated. The system successfully identified the anomaly and logged the event to Google Sheets within seconds, validating its real-time detection capability.

Index Terms: Cybersecurity, Risk Scoring, Artificial Intelligence, n8n, Automation.

I. INTRODUCTION

Relying solely on password verification during user authentication has become increasingly inadequate in the face of modern cyber threats. The real-time evaluation of login attempts originating from unfamiliar geographic locations, unusual time windows, or unrecognized devices has emerged as a critical security imperative.

Within the scope of this project, a node-based architecture that interconnects API services was preferred over writing heavyweight server-side code blocks. This design choice substantially reduced development time while enabling direct integration of artificial intelligence capabilities into the security pipeline. The study aims to demonstrate the practical viability of low-code automation frameworks for cybersecurity applications through a concrete implementation.

II. SYSTEM ARCHITECTURE AND INTEGRATION

The automation pipeline was constructed by connecting three primary components end-to-end within the n8n platform.

A. Data Collection via Webhook

The system's interface with the external world is established through a Webhook node that listens for incoming HTTP POST requests. The JSON payload, containing the fields User_ID, IP_Address, Location, and Device, is ingested and parsed by the system. During the testing phase, synthetic log entries were dispatched via Reqbin, confirming that the entry point correctly parsed all incoming parameters.

B. Artificial Intelligence (OpenAI) Analysis

In the second stage, the collected text data is forwarded directly to the OpenAI GPT-4o-mini module. The model is assigned a system prompt that defines its role as a cybersecurity expert and instructs it to score anomalies in the user's behavioral profile. To facilitate downstream processing, the model's output is constrained to a clean JSON structure containing only two fields: risk_score and reason, avoiding verbose free-form text.

C. Database Logging (Google Sheets)

Once risk assessment is complete, both the raw user data received from the Webhook and the score with its accompanying explanation generated by the AI are matched and appended as a new row to a Google Sheets spreadsheet. This step directly enhances the system's auditability and supports retrospective threat analysis.

III. RESULTS AND EXPERIMENTAL FINDINGS

Extreme scenarios were employed to validate the system's accuracy. Specifically, to evaluate the impossible travel condition, a synthetic request carrying a Chinese IP address was submitted to the system using an account profile that had previously logged in from Kastamonu, Turkey.

The AI module detected the abrupt geographic shift and the Unknown Android Device parameter, immediately assigning a risk score of 90, indicating high risk. The entire sequence—from HTTP request capture to row insertion in Google Sheets—completed within seconds, confirming the system's real-time logging capability.

These outcomes demonstrate that even low-code automation frameworks, when augmented with large language models, can construct enterprise-grade security layers. The system's flexible architecture can readily be extended to incorporate additional data sources, alternative AI models, and diverse notification channels such as email, SMS, or Slack.

IV. DISCUSSION

The proposed system transcends the rigidity inherent in rule-based approaches by incorporating contextual reasoning into the security workflow. Integrating large language models such as GPT-4o-mini into automation pipelines offers notable advantages over traditional classifiers, particularly in reducing false positive rates and providing human-interpretable decision rationales.

Nevertheless, LLM-based security systems introduce certain constraints. Model inference latency may generate delays in high-traffic environments, and minor variations in prompt engineering can directly affect scoring consistency. Additionally, the transmission of personal data such as user location and device information to a third-party API raises considerations that must be carefully assessed from a data privacy standpoint.

V. CONCLUSION

This paper presented the design and implementation of a cloud-based user access risk scoring system built on the integration of the n8n automation platform and the OpenAI GPT-4o-mini model. The system successfully unifies real-time detection of suspicious login attempts, AI-driven risk scoring, and persistent event logging into a cohesive pipeline.

Empirical tests confirmed that the system can identify complex anomalies—such as impossible travel scenarios—with high reliability. These findings support the conclusion that low-code AI integrations represent an accessible and effective alternative for modern Security Operations Centers (SOCs). Future work will focus on optimizing model inference latency and evaluating system performance under larger-scale, multi-layered threat scenarios.

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